

EXPERIMENT 18:

Perform Edge detection using Sobel Matrix along Y axis

AIM:- Edge detection using Sobel Matrix along y axis

PROGRAM:-

```
import cv2
import numpy as np
import os

image_path = r'C:\Users\pavan\OneDrive\OpenCV\mickey.png'
if not os.path.exists(image_path):
    print("Image file not found at:", image_path)
    exit()

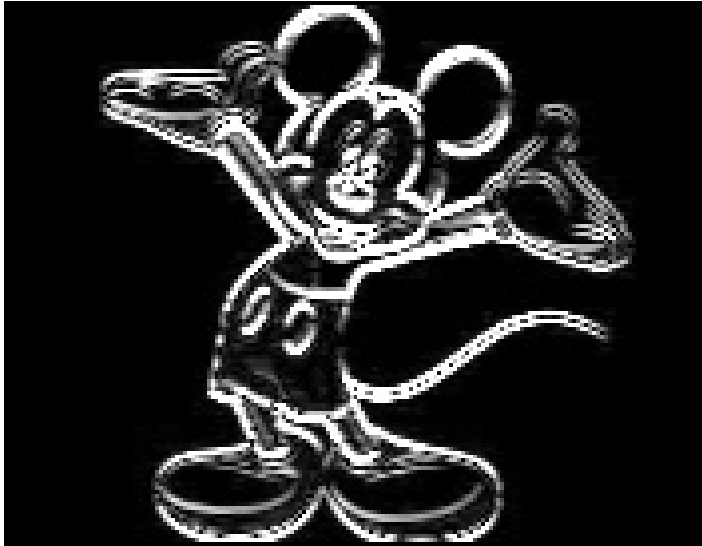
img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
if img is None:
    print("Failed to load image.")
    exit()

sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)
sobel_y = cv2.convertScaleAbs(sobel_y)
cv2.imwrite('SobelY.jpg', sobel_y)
cv2.imshow('Sobel Y-axis Edges', sobel_y)
cv2.waitKey(0)
cv2.destroyAllWindows()
print("Sobel Y-axis edge detection complete. Output saved as SobelY.jpg")
```

INPUT:-



Output:-



RESULT:-

Thus, the program is executed successfully